Fundamental of Computer Programming (FOCP)

BY

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Course Instructions

- Compulsory Attendance in Class (75%).
- Exams : Mid Term, End Sem CLAs (CLA: Continuous Learning Assessment)
- CLAs: Mini Test, Quizzes, Assignments, Group Discussion, Active Learning: Flipped Classes, Poster Presentation, Report, Problem/Project based Learning, Minute Paper, Viva (*No Late Submissions)
- Online Learning.
- Google/Teams Classroom.
- Demonstration of Lab work.

Distribution of Marks (Theory)

Continuous Evaluation (40 marks)					End Term Evaluation (60 marks)
Mid Term	Mini Test	Attendance	Class Performance	Assignments / Presentation / Active Learning	Written
20 marks			20 marks		60 marks

Distribution of Marks (Practical)

Continuous Evaluation (40 marks)			End Term Evaluation (60 marks)		
Attendance	Practical Experiment Results	Lab Performance	Written	Lab Exp. Demo	Viva
40 marks			15 marks	15 marks	30 marks

List of Textbooks

- Kernighan , B.W. and D.M.Ritchie , The C Programming Language , 2nd ed., Pearson Education India , 2015.
- Yashavant Kanetkar, Let Us C: Authentic Guide to C Programming Language (18th Edition) , BPB Publications,India,2021.
- KingK.N , C Programming :A Modern Approach , 2nded., W.W.Norton & Company, 2008.
- E.Bala guru swamy , Programming in ANSIC , Eighth edition , McGraw Hill Education (India) Private Limited , Noida , Uttar Pradesh , 2019.

Course Content

Module-1: Fundamentals of Programming

Brief History of Computing and Computers , Basic Organization of Computer , Representing Information as BitPatterns , NumberSystem , Basics of Computer Languages , Generation of Programming Languages , Compilers , Interpreter , Programming Environments and Debugging , Types of Errors and Debugging Techniques , Problem-SolvingAspects , Introduction to Algorithms , Flowcharts , Pseudocode.

Module-2: Basic Programming and Control Statements

StructureofCProgram,LifeCycle of Program from Source Code to Executable,Keywords,Identifiers,PrimitiveDataTypesinC, Variables,Constants, Input/Output Statements inC,Operators,TypeConversionandType Casting,Conditional Branching Statements , Iterative Statements , Nested Loops , Break and Continue Statements.

Module-3: Modular Programming and Recursion

Functions , Declaration , Definition , Call and Return , Call by Value, Call by Reference , Showcase Stack Usage with help of Debugger , Scope of Variables , Storage Classes , Recursive Functions , Recursion vs Iteration.

Module-4: Array-based Programming

Arrays , One-dimensional , Two Dimensional , and Multidimensional Arrays , Operations on Array , Traversal , Insertion , Deletion , Merging and Searching , Inter-Function Communication via Arrays , Passing a Row , Passing the Entire Array, Matrices , Strings , Read Operation , Writing and Manipulating Strings.

Module-5: CPointers , DMA and Structure

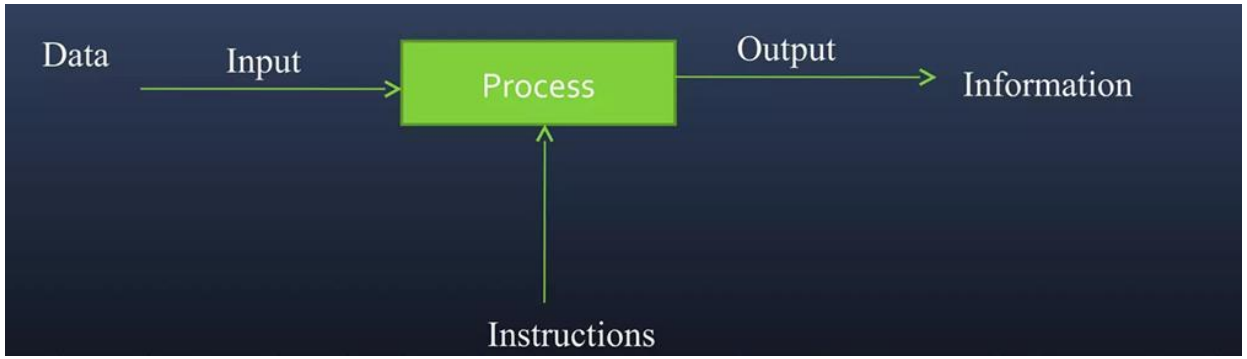
Pointers , Understanding Computer Memory, Accessing via Pointers , Pointers to Arrays , Drawback of Pointers , Dynamic Memory Allocation , Structures , Unions.

Module-6:File in C Programming

File Handling , File Redirection , File Pointers , Preprocessor , Library Functions , Low-Level Programming.

What is Computer?

- Derived from word **Compute.**
- A computer is an electronic device that takes in data (input), processes it according to programmed instructions (software), and produces results (output), while also storing information for later use.



History of Computing

Pre-1600 to 1800s	Early Computing Devices
1800s	Mechanical Era
1900 to 1940s	Electromechanical and Electronic Era
1940 to Till Now	Generations of Computer

Early Computing Devices

- **Abacus (c. 4000 BCE):** One of the earliest known tools for arithmetic, used in **ancient China and Mesopotamia.** Used beads sliding on rods to perform arithmetic operations. Still used today in some parts of Asia for teaching math and mental calculation.
- **Napier's Bones (1617):** Invented by **John Napier** to simplify multiplication and division. Consisted of rods inscribed with multiplication tables—users aligned them to calculate products. Based on lattice multiplication, it was a precursor to mechanical calculators.
- **Pascaline (1642):** A mechanical calculator by **Blaise Pascal** that could perform addition and subtraction using a series of gears and wheels. One of the first mechanical calculators to be mass-produced.
- **Leibniz's Stepped Reckoner (1673):** Improved on Pascal's design to perform all four basic operations. Used a stepped drum mechanism to perform all four arithmetic operations.

Mechanical Era

❖ **Jacquard Loom (1804)**

- Used punch cards to control weaving patterns in textiles.
- Considered the first programmable machine.

❖ **Difference Engine (1822)**

- Designed by Charles Babbage to automate polynomial calculations.
- Could compute and print mathematical tables.

❖ **Analytical Engine (1837)**

- Also, by Babbage, it was the first design for a general-purpose computer.
- Included components like a “mill” (CPU), “store” (memory), and punch card input/output
- Ada Lovelace wrote the first algorithm for it, making her the world’s first programmer.

Electromechanical and Electronic Era

❖ **Tabulating Machine (1890)**

- Invented by Herman Hollerith for the U.S. Census.
- Used punch cards and electrical circuits to tabulate data.
- Led to the founding of IBM.

❖ **Z3 (1941)**

- Built by Konrad Zuse in Germany.
- First programmable, fully automatic digital computer.
- Used electromechanical relays and binary arithmetic.

❖ **ENIAC (1945)**

- First general-purpose electronic computer.
- Used vacuum tubes and could perform 5,000 additions per second.
- Required rewiring to change programs—no stored-program capability.

Generation of Computers

Generation Number	Time duration	Technology
1st	1940-1956	Vacuum Tubes
2nd	1956-1963	Transistors
3rd	1964-1971	Integrated Circuits
4th	1971-2010	Microprocessors
5th	2010-Present	AI & Quantum Tech

First Generation: Vacuum Tube Computers

Technology	**Vacuum tubes for circuitry, magnetic drums for memory.**
Programming	**Machine language (binary code).**
Size & Power:	**Massive machines that consumed a lot of electricity and generated heat.**
Examples:	**ENIAC, UNIVAC I, IBM 701.**
Speed:	**Very slow—measured in milliseconds.**

Conclusion: Limited memory space and Processing Power, required Air Conditioning.

Second Generation: Transistor Computers

Technology	**Transistors replaced vacuum tubes—smaller, more reliable, and energy-efficient.**
Programming	**Assembly language and early high-level languages like COBOL and FORTRAN.**
Size & Power:	**Smaller and more efficient than first-gen.**
Examples:	**IBM 1401, CDC 1604**
Speed:	**Microseconds.**

Conclusion: Used in business, universities, and government; introduced batch processing & multiprogramming.

Third Generation: Integrated Circuit (IC) Computers

Technology	**Integrated Circuits (ICs) combined multiple transistors on a single chip.**
Programming	**High-level languages like BASIC, Pascal.**
Size & Power:	**Even smaller, more reliable, and cheaper.**
Examples:	**IBM System/360, PDP-8.**
Speed:	**Nanoseconds.**

Conclusion: Introduced operating systems, keyboards, monitors, and time-sharing.

Fourth Generation: Microprocessor Computers

Technology	**Microprocessors (entire CPU on a single chip).**
Programming	**Advanced high-level languages (C, C++, Java).**
Size & Power:	**Desktop and personal computers became common.**
Examples:	**Intel 4004, Apple Macintosh, IBM PC.**
Speed:	**Picoseconds.**

Conclusion: Rise of personal computing, graphical user interfaces (GUIs), and the internet.

Fifth Generation: Artificial Intelligence & Beyond

Technology	**AI, machine learning, quantum computing, cloud computing.**
Programming	**Natural language processing, neural networks, Python, R.**
Size & Power:	**Ultra-portable (smartphones, wearables), cloud-based infrastructure..**
Examples:	**IBM Watson, Google DeepMind, quantum prototypes.**
Speed:	**Real-time processing, parallel computing.**

Conclusion:Automation, self-learning systems.

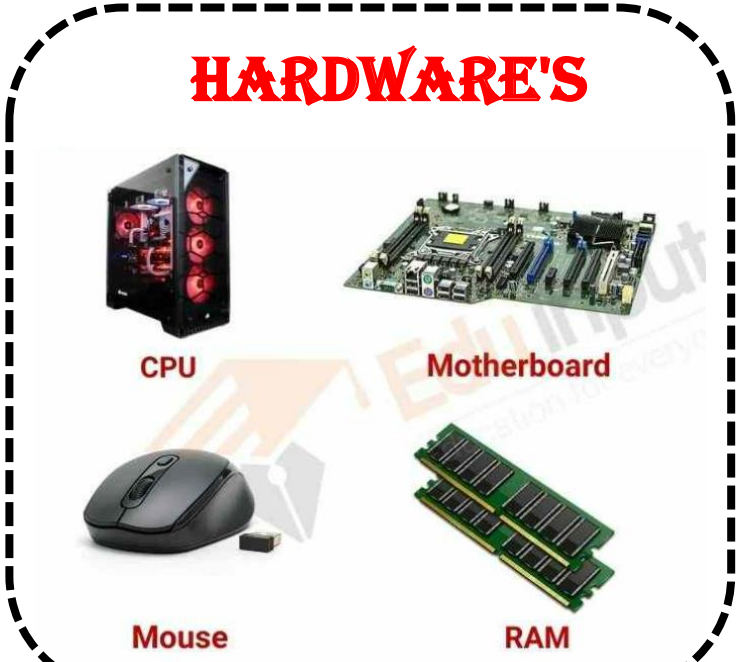
Fundamentals of Computer

SOFTWARE'S





HARDWARE'S



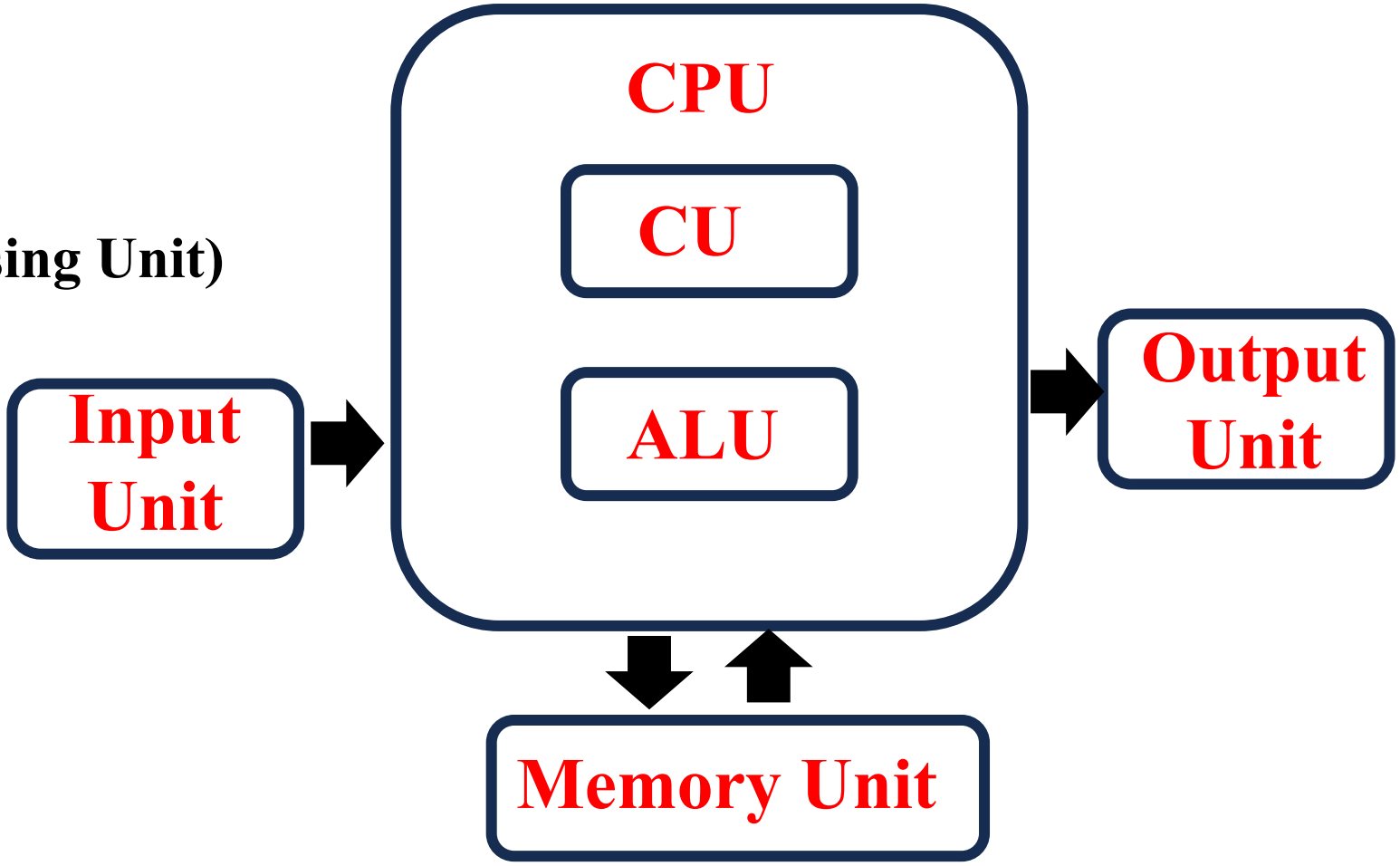
- **Hardware: Physical Access to the parts of Computer**
- **Software: Set of Instruction for the Computer to perform the task.**

Fundamentals of Computer Organization

❖ **Computer Organization: This states the working of the internal parts of the computer (e.g. Working of ALU (Arithmetic Logic Unit), Memory Unit)**

Every Computer has 4 Units

1. **Memory Unit**
2. **Input Unit**
3. **Output Unit**
4. **Processing Unit (Central Processing Unit)**



CU: Control Unit

CPU: Central Processing Unit

ALU: Arithmetic Logic Unit

Roles of Computer Units and Their Links

CPU: Coordinates with the memory unit to perform desired tasks.

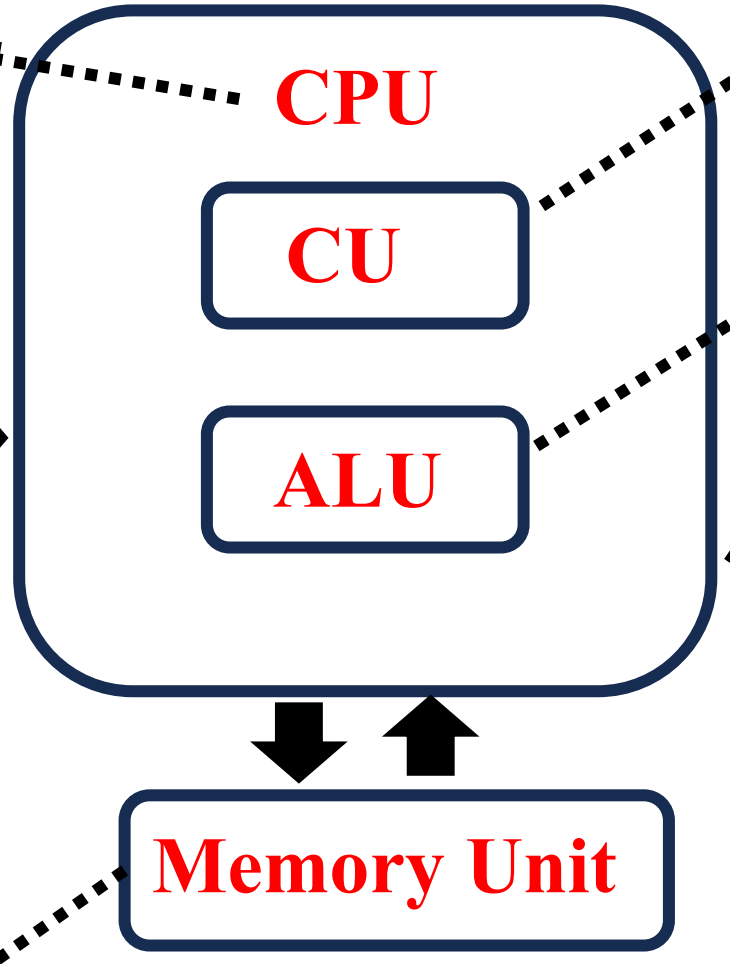
CU: Surveillances all the activities of CPU.

ALU: Performs all the logical operations and gives output to output unit.

Input Unit: A path through which a user interacts with the Processing Unit.

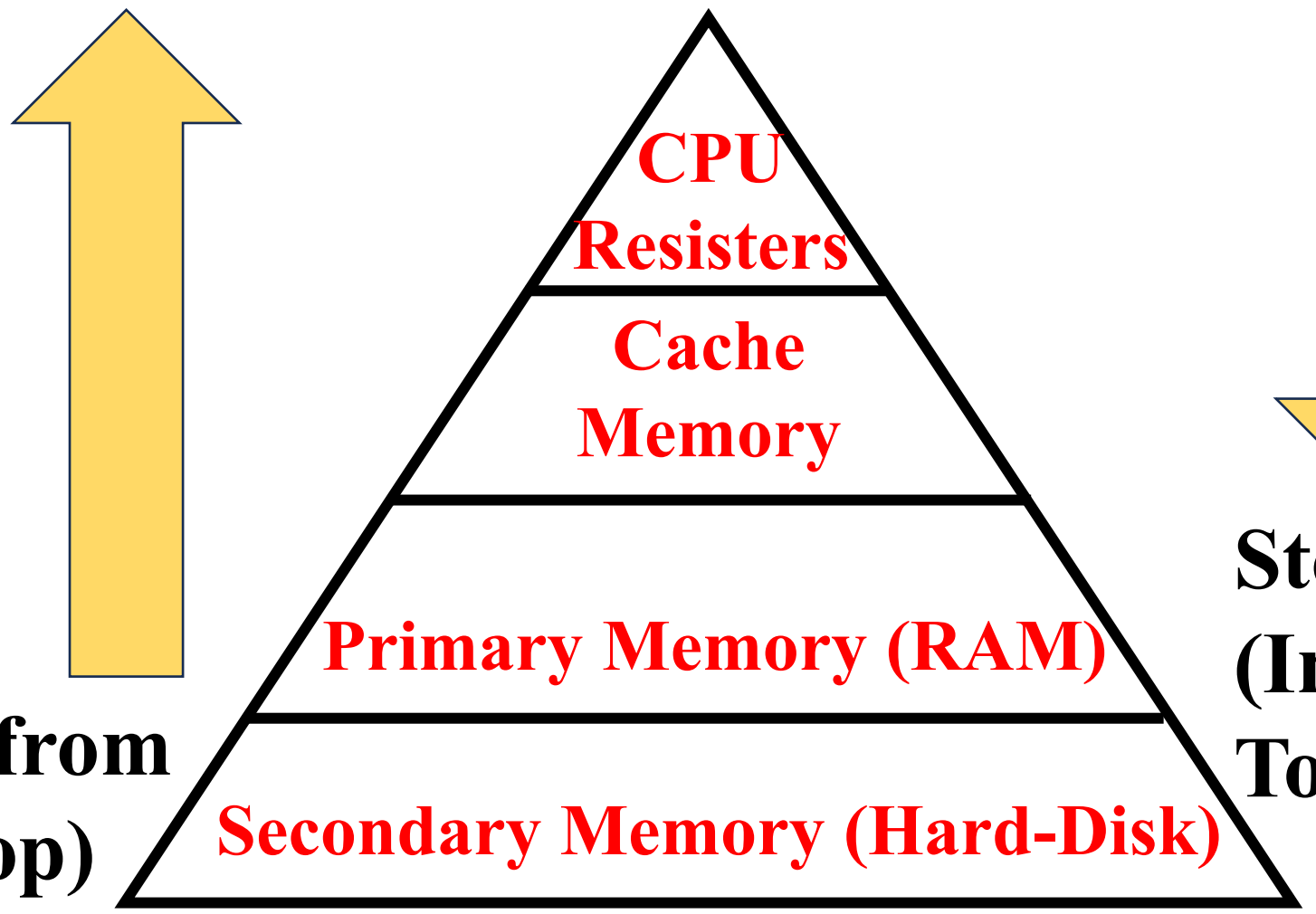
Memory Unit: Storage unit of the device.

Output Unit: A path through which processing unit interacts with the user.



Block Diagram of Computer Units

Memory Hierarchy

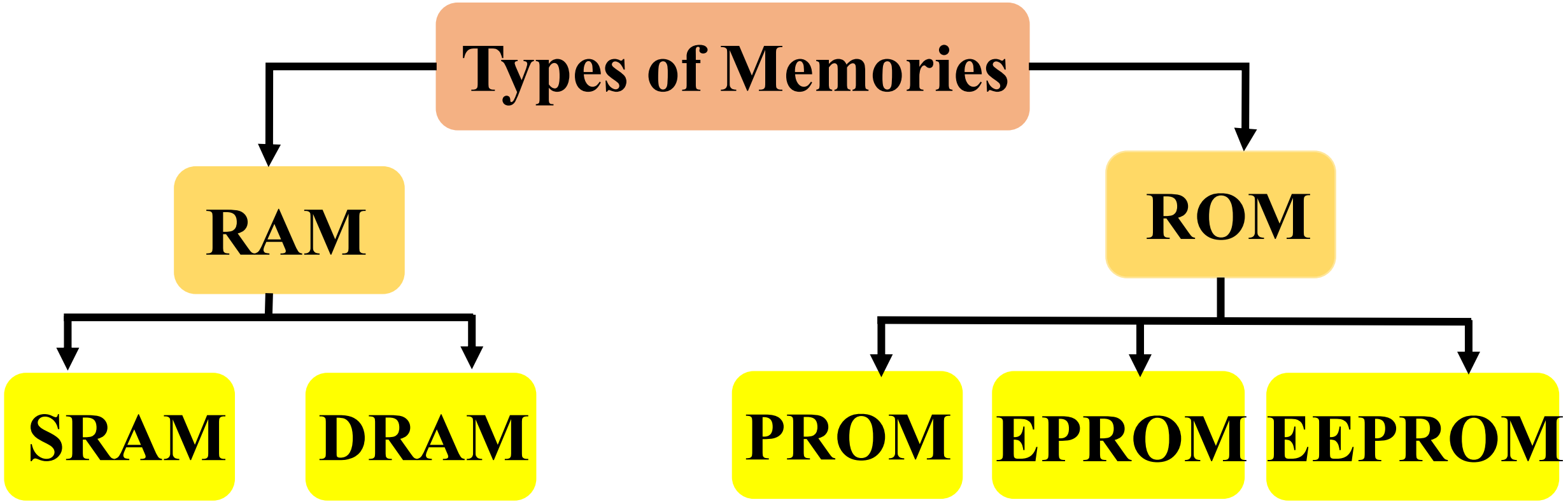


Speed (Increases from Down to Top)

Storage Capacity (Increases from Top to Down)

- **Secondary Memory: External Memory**
- **Primary Memory, Cache Memory, CPU Registers: Internal Memory**

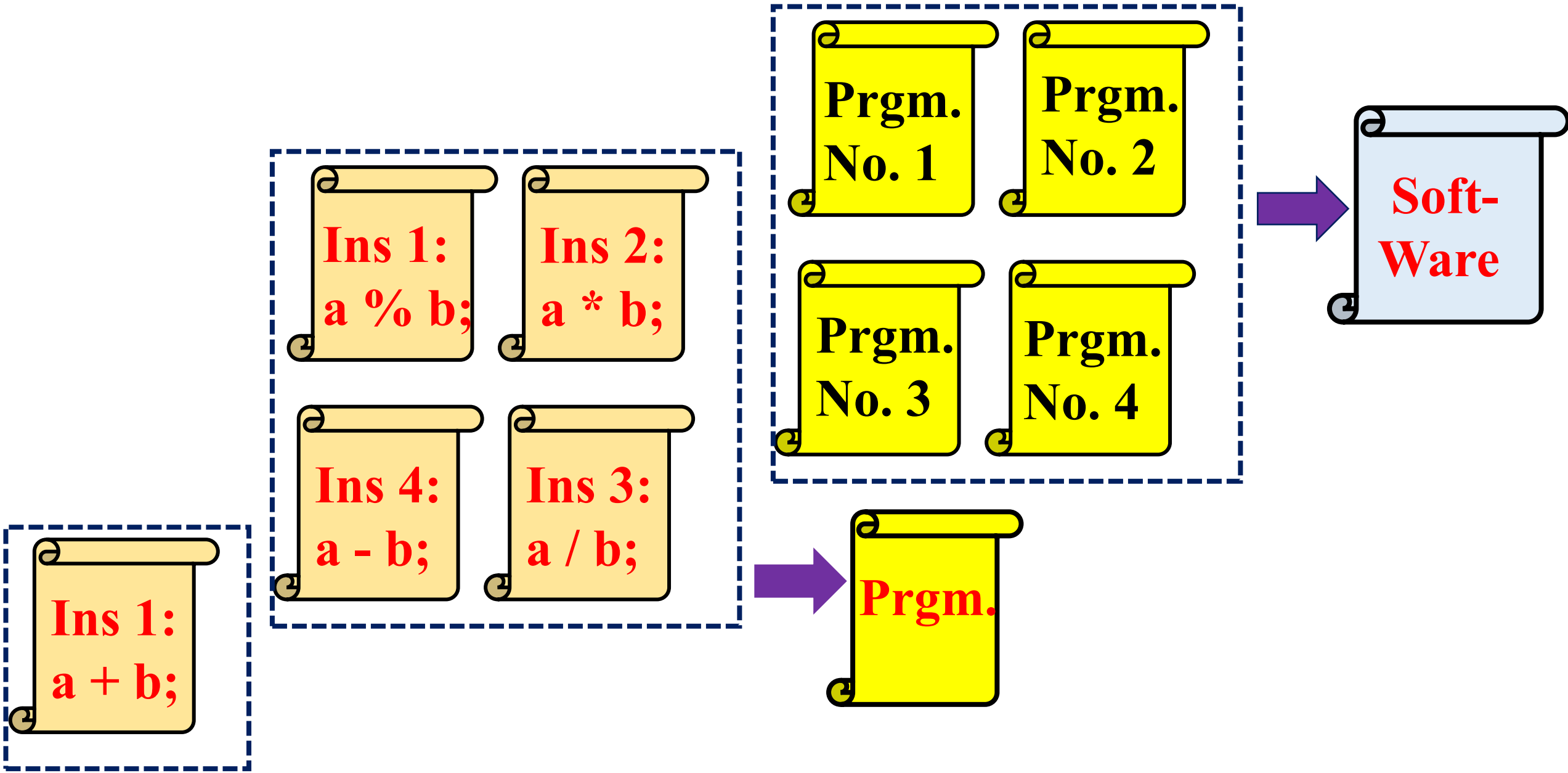
Memory Classification



- **SRAM: Static Random Access Memory**
- **DRAM: Dynamic Random Access Memory**

- **PROM: Programmable Random Only Memory**
- **EPROM: Erasable Programmable Random Only Memory**
- **EEPROM: Electrical Erasable Programmable Random Only Memory**

Software: Program: Instructions

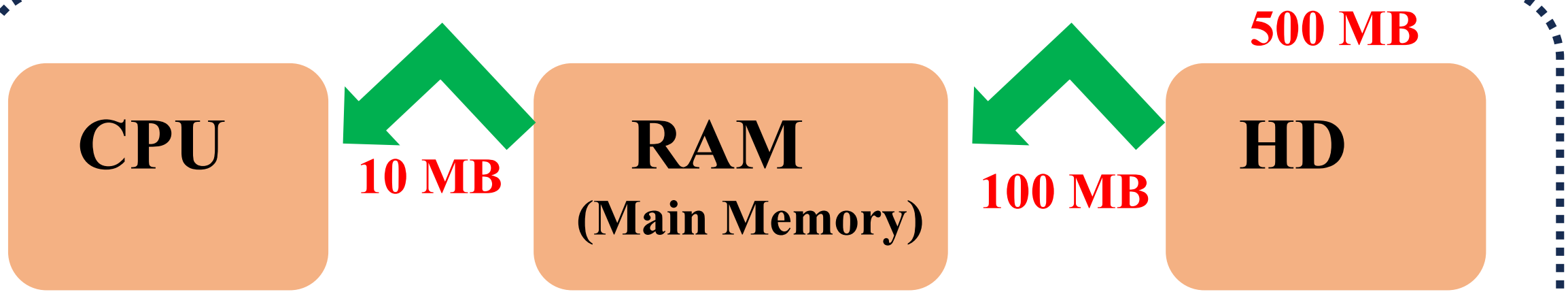


Instruction Cycle

For Instruction Execution

CPU Running any Software

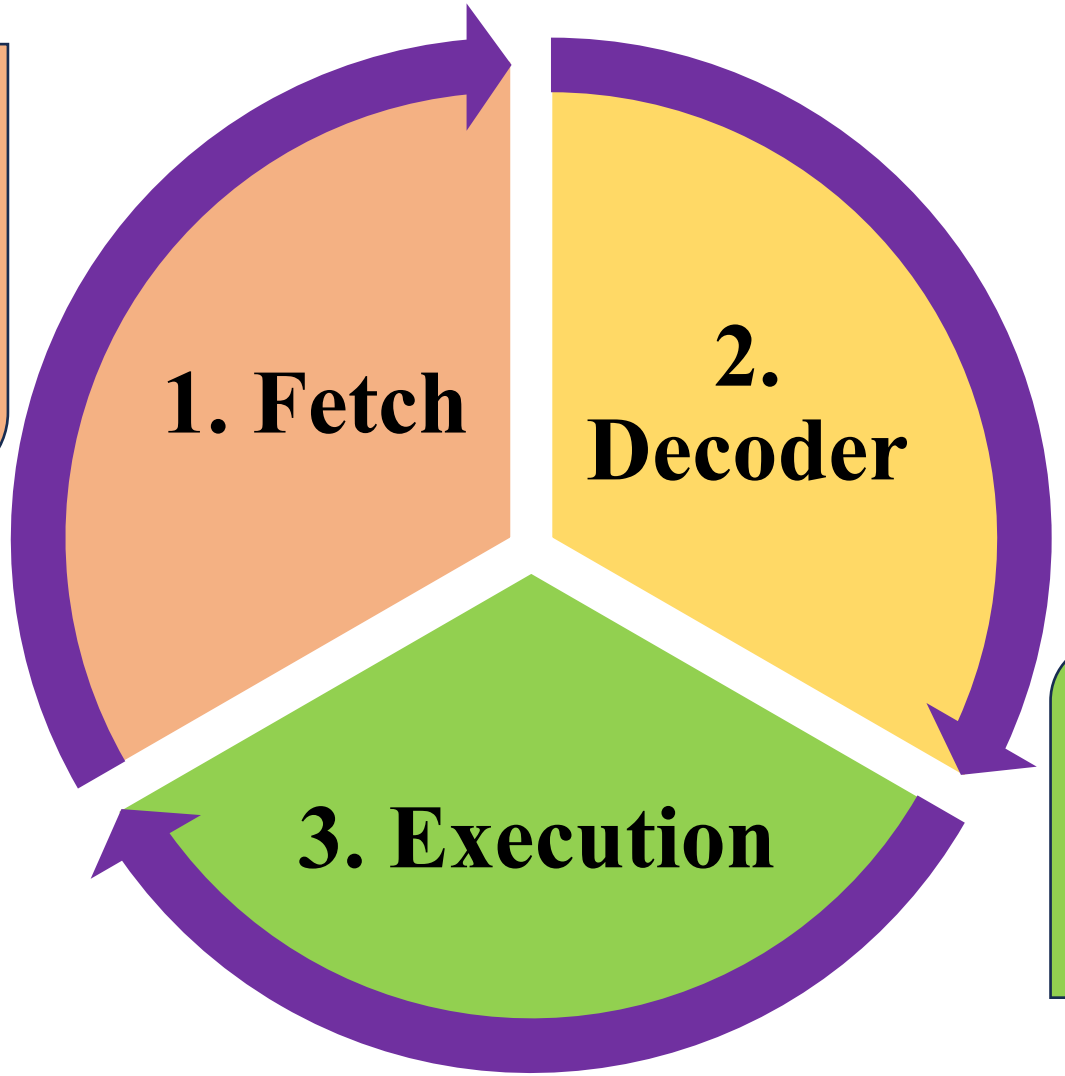
1. CPU takes data from Hard Disk (HD).
2. Data Transferred from HD to Main Memory (RAM).
3. CPU takes only necessary data which is required.



Block Diagram of Execution of any Instruction

Instruction Cycle

1. Fetch: Instruction is fetched from HD to RAM and RAM to CPU.



2. Decode: Instruction is converted into ML or BL. (Machine language, Binary Language)

3. Execution: Allotted function is performed.

Ex. 1 program : 12 Instructions.
Software has 10 programs
120 Instructions
After 120 Cycles desired output would be obtained.

Performance Equation

$$T = \frac{N \times S}{R}$$

T: Time required by the processor to execute single program.
N: Number of Instructions.
S: Steps involved in every instructions.
R: Processor speed (Cycle/seconds)

Representing Information as Bit Patterns

What is a Bit?

A **bit** (short for **binary digit**) is the **smallest unit of data** in a computer.
It can have only **two possible values**:

- **0 (OFF/False)**
- **1 (ON/True)**

All digital data — numbers, text, images, audio — is ultimately represented using combinations of **0s and 1s**, known as **bit patterns**.

Why Bit Patterns?

- Computers use **binary (base-2)** number system.
- Electrical circuits in a computer can easily represent two states:
  - High voltage (1)
  - Low voltage (0)
- Therefore, all information must be **converted into binary format**.

Types of Data Represented Using Bit Patterns

- ❑ Numbers (Integers and Floating-point)
- ❑ Characters (Text)
- ❑ Images (Pixels and Colors)
- ❑ Sound (Audio Samples)
- ❑ Video (Sequence of images + audio)

Number System

- In digital system, the number system is used for representing the information.
- The number system has different bases.
- The base represent as the total number of digits used in the number system.
- There are four types of number systems.

Number System

- **Binary System: 0 and 1 (2 Numbers)**
- **Octal System: 0 to 7 (8 Numbers)**
- **Decimal System: 0 to 9 (10 Numbers)**
- **Hexadecimal System: 0 to 9, A,B,C,D,E and F (16 Numbers)**

Where A: 10 , B:11 , C:12, D:13, E:14, F:15

Number Representation

Format of all types of Systems: $(a_3a_2a_1a_0)_b$

e.g. $(245)_8$, $(CAC)_H$, $(1001)_2$, etc.

"b": Base (for Binary:2, Oct:8) (a:0,1 and a: 0 to 7)

- **"b >a" Always and Always but "a cannot be > b"**



MSB: Most Significant bit

LSB: Least Significant bit

How to expand this?

$$(a_3a_2a_1a_0)_b = (b^3 \times a_3) + (b^2 \times a_2) + (b^1 \times a_1) + (b^0 \times a_0)$$

$(8245)_{10}$: $(8000+200+40+5) = (8 \times 10^3) + (2 \times 10^2) + (4 \times 10^1) + (5 \times 10^0)$

Binary to Decimal Conversion

For Numbers After Decimals: $(a_0.a_{-1}a_{-2}a_{-3})_b$

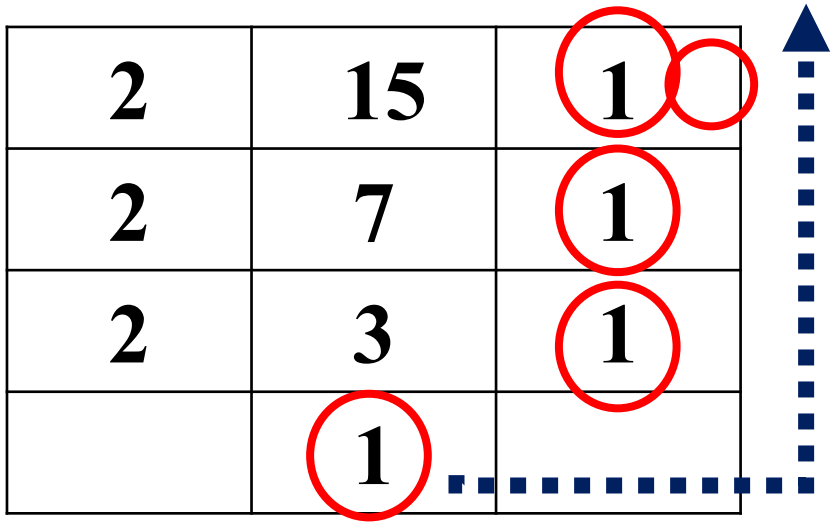
How to expand this $(a_0.a_{-1}a_{-2}a_{-3})_b = (b^0 \times a_0) + (b^{-1} \times a_{-1}) + (b^{-2} \times a_{-2}) + (b^{-3} \times a_{-3})$

- $(1001)_2 = (2^3 \times 1) + (2^2 \times 0) + (2^1 \times 0) + (2^0 \times 1) = 8 + 1 = 9$
- $(111.1)_2 = (2^2 \times 1) + (2^1 \times 1) + (2^0 \times 1) + (2^{-1} \times 1) = 7 + 0.5 = 7.5$
- $(11000)_2 = ?$
- $(0101)_2 = ?$
- $(1110.01)_2 = ?$
- $(10010)_2 = ?$

Decimal to Binary Conversion

Division Approach is Used for Performing the Conversion

$$(15)_{10} = (?)_2$$



2	15	1
2	7	1
2	3	1
	1	

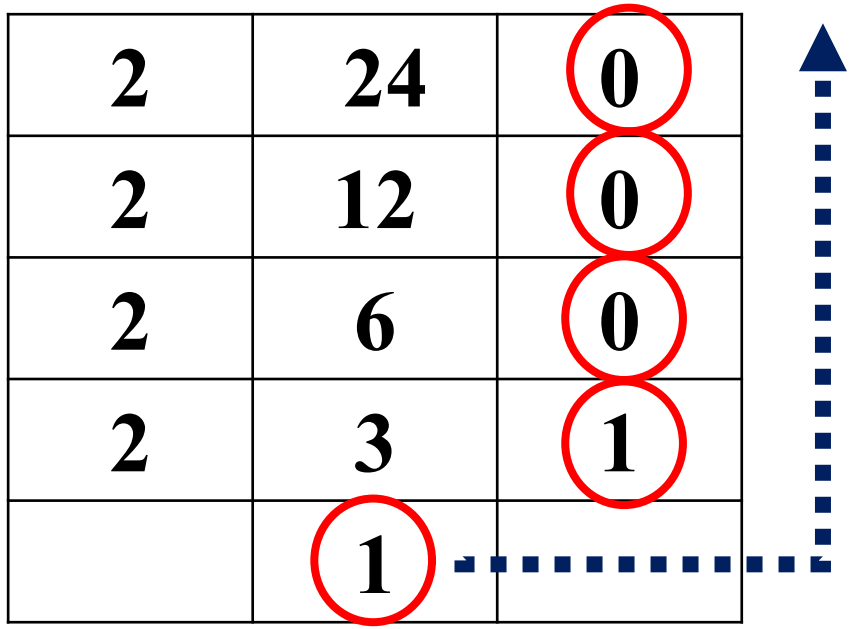
$$(1111)_2$$

Bottommost: MSB
Topmost: LSB

- **Step 1: Divide Number By 2 write quotient below and remainder aside**
- **Step 2: Perform the division until quotient is less than 2 is achieved**
- **Step 3: Write the Number from Bottom to Top**

Decimal to Binary Conversion

$(24)_{10} = (?)_2$



2	24	0
2	12	0
2	6	0
2	3	1
	1	

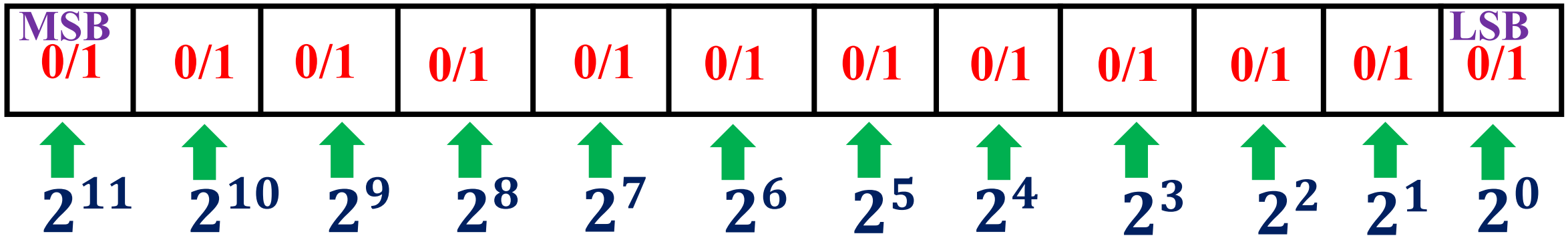
Ans:$(11000)_2$

$(32)_{10} = (?)_2$
$(64)_{10} = (?)_2$
$(18)_{10} = (?)_2$
$(56)_{10} = (?)_2$

$(31)_{10} = (?)_2$
$(63)_{10} = (?)_2$
$(15)_{10} = (?)_2$

Things to Remember (Power's of 2)

2^1	2	2^6	64
2^2	4	2^7	128
2^3	8	2^8	256
2^4	16	2^9	512
2^5	32	2^{10}	1024



Octal Number System

Numbering Sequence: 0,1,2,3,4,5,6,7,10,11,12,13,14,15,16,17,20,21,……

$(10)_8 = 8 \times 1 + 0 \times 1 = 8$

$(11)_8 = 8 \times 1 + 1 \times 1 = 9$

$(12)_8 = 8 \times 1 + 2 \times 1 = 10$

$(17)_8 = 8 \times 1 + 7 \times 1 = 15$

$(27)_8 = 8 \times 2 + 7 \times 1 = 23$

$(47)_8 = 8 \times 4 + 7 \times 1 = 39$

Octal to Decimal Conversion

$$(a_3a_2a_1a_0)_b = (b^3 \times a_3) + (b^2 \times a_2) + (b^1 \times a_1) + (b^0 \times a_0)$$

Decimal to Octal Conversion

8	62	6
	7	

$(62)_{10} = (76)_8$

8	600	0
8	75	3
8	9	1
	1	

8	285	5
8	35	3
	4	

$(285)_{10} = (435)_8$

$(600)_{10} = (1130)_8$

Conversion of Binary to Octal Number System

Make group of 3 , 3 bits starting from LSB

$$(1110011110000011111)_2$$

Grouping of 3 bits: $(111\ 001\ 111\ 000\ 001\ 111)_2$

Corresponding Octal Value: 7 1 6 0 3 7

Final Answer: $(716037)_8$

Hexadecimal Number System

Numbering Sequence: 0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F,10,11,12,13,……

$(10)_H = 16 \times 1 + 0 \times 1 = 16$

$(25)_H = 16 \times 2 + 5 \times 1 = 37$

$(AB)_H = 16 \times 10 + 11 \times 1 = 171$

$(A4)_H = 16 \times 10 + 4 \times 1 = 164$

$(DC)_H = 16 \times 13 + 12 \times 1 = 220$

Hexadecimal to Decimal Conversion

Decimal to Hexadecimal Conversion

16	62	E
	3	

$(62)_{10} = (3E)_H$

16	285	D
16	17	1
	1	

$(285)_{10} = (11D)_H$

16	600	8
16	37	5
	2	

$(600)_{10} = (258)_H$

Conversion of Binary to Hexa. Number System

Make group of 4 , 4 bits starting from LSB

$$(1110011110000011111)_2$$

Grouping of 4 bits: $(1110011110000011111)_2$

Corresponding Hex. Value: 3 9 C 1 F

Final Answer: $(39C1F)_H$